

Aim – Demonstrate GUI using Tkinter in python.

Theory – Graphical User Interface (GUI) programming in Python can be done using the Tkinter library. Tkinter provides tools to create windows, labels, buttons, and input fields. In this program, a student form interface is created where users enter details like name, roll number, branch, and email. The program retrieves input values from entry widgets and validates them. Message boxes are used to display warnings for incomplete data and confirmations when the form is successfully submitted.

A1.

```
import tkinter as tk

from tkinter import messagebox

def submit_form():
    name = entry_name.get()
    roll = entry_roll.get()
    branch = entry_branch.get()
    email = entry_email.get()
    if not (name and roll and branch and email):
        messagebox.showwarning("Incomplete Data", "Please fill all fields.")
    return
    messagebox.showinfo("Submitted", f"Name: {name}\nRoll No: {roll}\nBranch: {branch}\nEmail: {email}")

root = tk.Tk()
root.title("Student Form")
root.geometry("300x300")

tk.Label(root, text="Student Form", font=("Arial", 16)).pack(pady=10)

tk.Label(root, text="Name:").pack()
entry_name = tk.Entry(root)
entry_name.pack()

tk.Label(root, text="Roll No:").pack()
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```
entry_roll = tk.Entry(root)
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entry_roll.pack()
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```
tk.Label(root, text="Branch:").pack()
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```
entry_branch = tk.Entry(root)
```

```
entry_branch.pack()
```

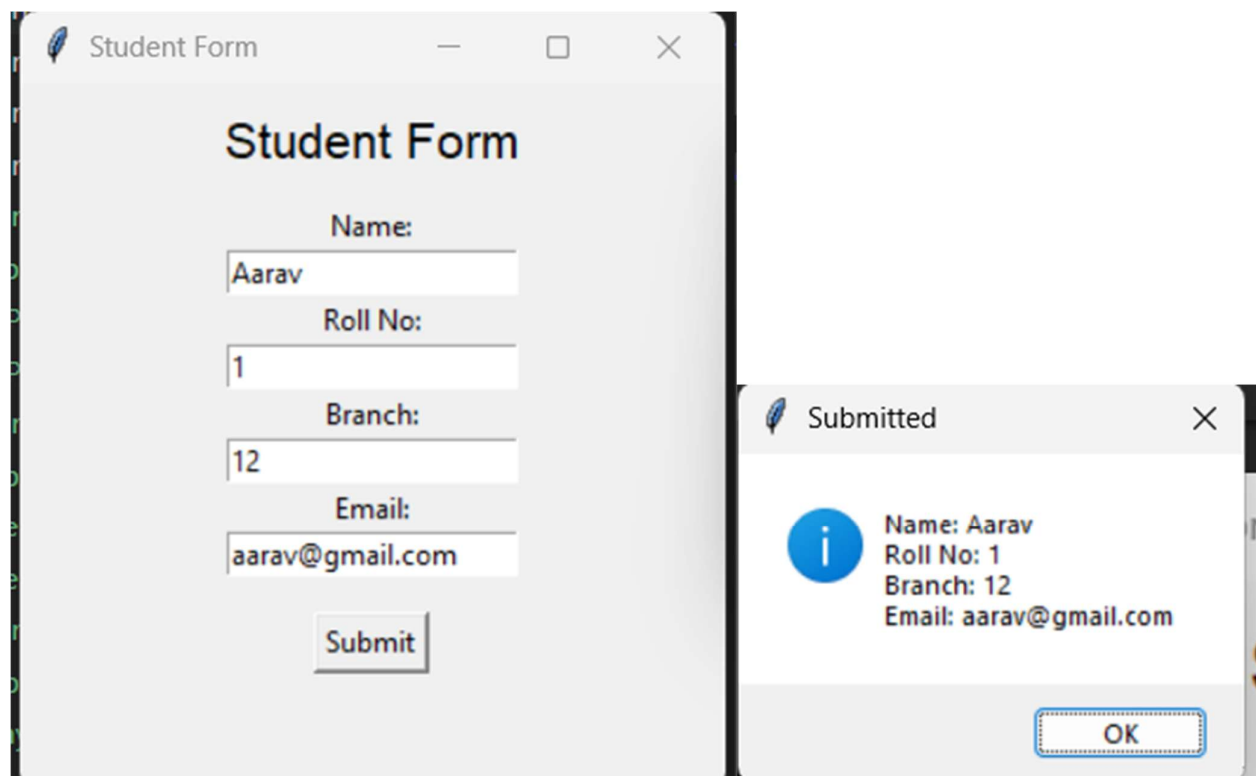
```
tk.Label(root, text="Email:").pack()
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```
entry_email = tk.Entry(root)
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```
entry_email.pack()
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```
tk.Button(root, text="Submit", command=submit_form).pack(pady=15)
```

```
root.mainloop()
```



Conclusion

This program demonstrates how Tkinter can create simple GUI forms in Python, allowing user input, basic validation, and interactive feedback through message boxes in a user-friendly interface.